

MOBILE APPLICATION DEVELOPMENT ASSIGNMENT 1

**Multi-Activity App with Intents and Menus
Student Task Planner | TaskFlow**

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**Course: CIT-646 Mobile Application
Development**

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TaskFlow Documentation

Application Overview

1. Introduction

TaskFlow is a multi-screen Android application designed to help students organize and manage their academic tasks. The application allows users to create tasks, view task details, mark tasks as completed, delete tasks, sort tasks, and share task information. The application was developed using Kotlin and XML in Android Studio.

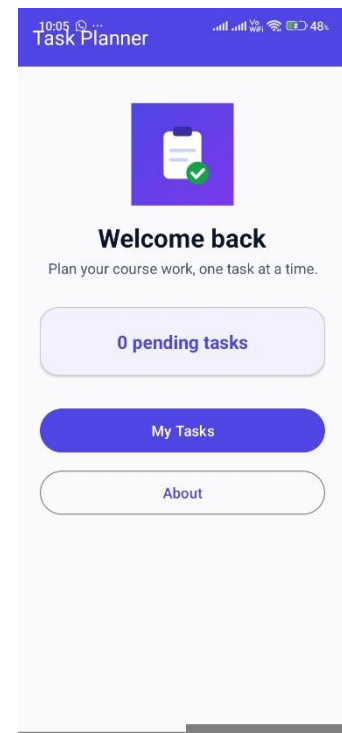
2. Technologies Used

Technology	Purpose
Android Studio	Application development
Kotlin	Application logic
XML	User interfaces
Android Intents	Activity navigation and external actions
RecyclerView/ListView	Task list
Menus	User actions
Activity Result API	Returning task data/results

Application Screens

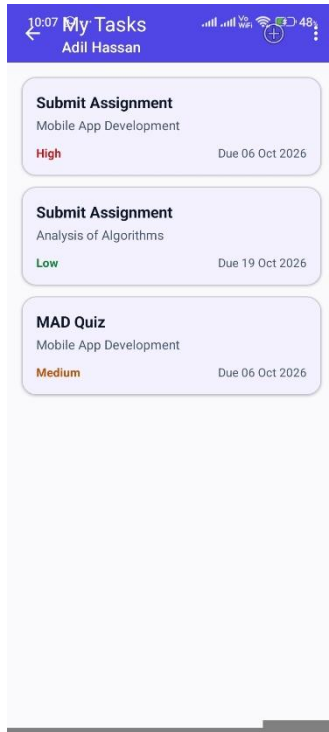
a. Dashboard

A simple Todo Application developed by Adil Hassan. This application developed in Android Studio. It saves to-dos in-memory, not in database just now. You can set title, priority and due date. Also, you can see details of your task. You can sort your to-dos with some filters.



b. Task List

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In this screen you can see all your tasks. You can sort them by due date, priority and some other ways. By clicking you can see task details also can delete it from task details.

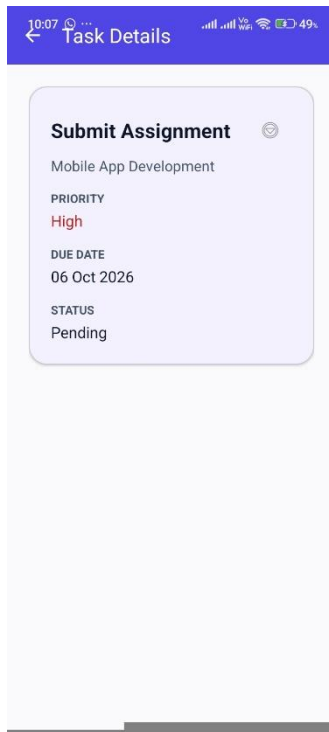
c. Add Task

A screenshot of a mobile application interface titled 'Add Task'. It features four input fields: 'Task title', 'Course', and 'Due date'. Below these fields is a 'Priority' section with three radio buttons: 'Low', 'Medium' (which is selected), and 'High'. At the bottom, there are two buttons: 'Cancel' and 'Save'. The top status bar shows the time as 10:40, signal strength, Wi-Fi, and 60% battery.

From this screen you can create a new task. All things are compulsory and must be filled, without this task will not create. All things choose with caution.

d. Task Details

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Here you can see your task details. From here you can share, mark done and delete your task.

e. About

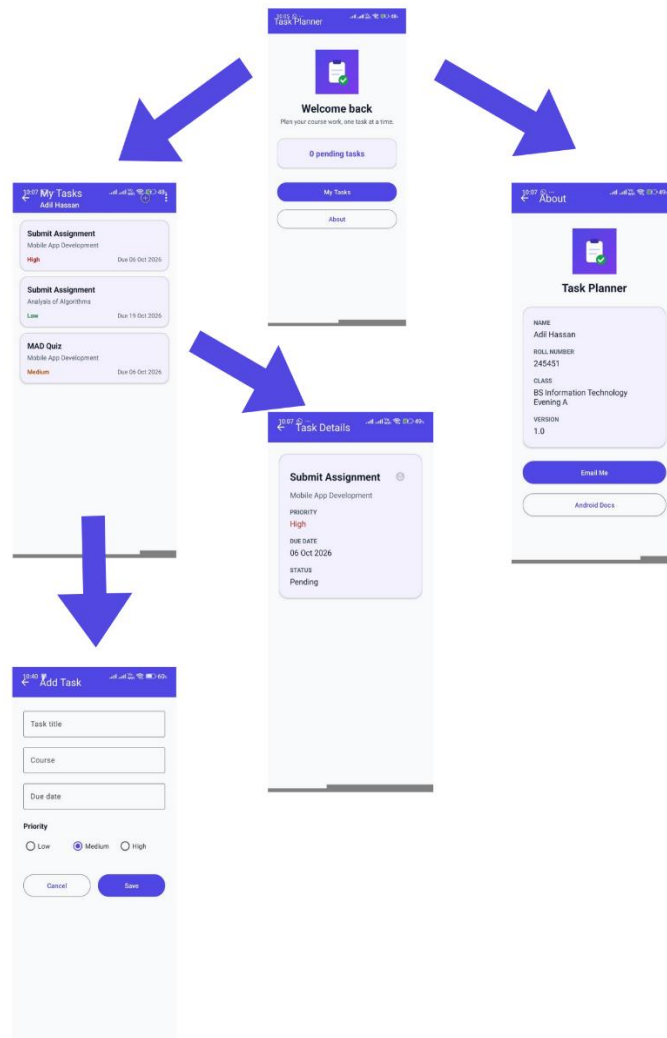


This screen provides information about the application, its version, creator and some others. This screen gives you details. You can send me email from here also you can check name, Android Docs.

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Intent & Navigation

ID	Navigation	Intent	Purpose
N1	Dashboard → Task List	Explicit	Opens tasks and passes student name
N2	Dashboard → About	Explicit	Opens About
N3	Task List → Add Task	Explicit + Result	Adds/returns task data
N4	Task List → Task Detail	Explicit	Send task information
N5	Task Detail → Task List	Result	Returns DONE/DELETE action
N6	Task Detail → Share	Implicit	Shares task summary
N7	About → Email	Implicit	Opens email application
N8	About → Browser	Implicit	Opens Android documentation



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Menu Implementation

Explain the three menu types.

Menu options

Used on the Task List screen:

Add

Sort

- |— By Priority
- |— By Due Date
- |— By Title

About

Exit

Context Menu

Activated by long pressing a task:

View Details

Mark Done

Delete

Popup Menu

Available from the three-dot button on Task Details:

Share

Mark Done

Delete

CLO-1 Questions

Q1. When TaskListActivity starts AddTaskActivity, which life-cycle callbacks run in each activity, and in what order?

Answer: When TaskListActivity starts AddTaskActivity, the current activity first goes through onPause() and then onStop() as it leaves the foreground. After that, AddTaskActivity goes through onCreate(), onStart(), and onResume(). Finally, AddTaskActivity becomes the active and interactive activity.

Q2. What happens to the back stack when the user presses Back on TaskDetailActivity, and what happens when the Exit menu item calls finishAffinity()?

Answer: When the user presses Back on TaskDetailActivity, the current activity is removed from the top of the back stack, and the user returns to TaskListActivity. However, when the Exit menu calls finishAffinity(), it finishes the activities associated with the current task, effectively closing the app's activity stack.

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Q3. Why do N1 to N5 use explicit Intents while N6 to N8 use implicit Intents? What happens if no app can handle an implicit Intent, and how does your app respond?

Answer: N1 to N5 use explicit Intents because they are used for navigation between specific activities within our own TaskFlow application, so the exact destination activity is known. N6 to N8 use implicit Intents because they request actions that can be handled by other applications, such as sharing content, sending an email, or opening a web browser. If no application can handle an implicit Intent, Android throws an `ActivityNotFoundException`; our app catches this exception and displays a Toast message instead of crashing.

Q4. Name three user-interface principles you applied and give one example of each from your app.

Answer: Consistency: I used consistent colors, typography, spacing, and button styles throughout the app. For example, the same Material 3 theme and color scheme are used across the dashboard and task screens.

Simplicity: I kept the interface clean and focused on the main task-management functions. For example, the dashboard clearly displays My Tasks, Pending Tasks, and the Add Task action without unnecessary elements.

Visibility & Feedback: The app provides clear visual information about task status and priority. For example, tasks use different priority indicators for High, Medium, and Low, while completed tasks can be marked as Done.

Testing & Conclusion

Test	Expected Result	Status
Open My Tasks	Task List opens	✓
Add valid task	Task appears in list	✓
Save empty title	Validation error	✓
Cancel Add Task	Returns safely	✓
Open task	Details displayed	✓
Mark Done	Task becomes completed	✓
Delete task	Task removed	✓
Share task	Share app opens	✓
Email Me	Email app opens	✓
Android Docs	Browser opens	✓
Long press task	Context menu opens	✓
Three-dot button	Popup menu opens	✓
Sort tasks	List reordered	✓

Conclusion

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TaskFlow successfully demonstrates the use of multiple Android activities, explicit and implicit Intents, Activity Result API, Options Menu, Context Menu, and Popup Menu. The application provides a consistent and responsive user interface for managing student tasks while demonstrating navigation, data passing, event handling, and user feedback.